A Texture-Enhanced Deep Learning Pipeline for Binary Class Cervical Cytology Classification using the SIPaKMeD Dataset

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Dataset Used: SIPaKMeD (Pap Smear Dataset)
Link: https://www.cs.uoi.gr/~marina/sipakmed.html
Description: 4049 Pap smear cell images across five classes. This study uses only Metaplastic and Koilocytotic classes for binary classification.

I. Introduction

Cervical cancer is one of the most common cancers among women, and early diagnosis is very important for successful treatment. One of the most widely used screening methods is the Pap smear test, where cervical cells are examined under a microscope to detect abnormal changes [1], [2]. However, examining these cell images manually is a difficult and time-consuming process. In addition, the results may sometimes depend on the experience of the cytopathologist, which can lead to differences in diagnosis [2].

One of the main challenges in Pap smear analysis is that some cervical cell types look very similar to each other. In particular, Metaplastic cells, which are benign, and Koilocytotic cells, which are abnormal, may have similar nucleus structures, staining patterns, and overall morphology. Because of these similarities, distinguishing these two cell types correctly can become difficult even for experienced specialists.

In this project, a deep learning-based classification system is proposed to distinguish between Metaplastic and Koilocytotic cervical cells. The problem is designed as a binary image classification task, and the main goal is to improve the classification performance between these two visually similar cell types.

The proposed system will use transfer learning with the ResNet-50 model [3]. In addition, several preprocessing and data augmentation techniques will be applied to improve the quality of the training process. Grad-CAM visualizations [4] may also be used to better understand which image regions influence the model during classification.

II. Problem Definition & Motivation

Cervical cancer screening is one of the most important procedures for the early detection of precancerous abnormalities. In Pap smear analysis, cytopathologists examine cervical cell images under a microscope and determine whether the observed cells are normal, benign, or abnormal. However, this process is difficult, time-consuming, and highly dependent on expert interpretation.

One of the most challenging problems in cervical cytology is the morphological ambiguity between certain cell types. In particular, benign Metaplastic cells often exhibit visual characteristics that are very similar to abnormal Koilocytotic cells. Their nucleus appearance, chromatin texture, staining distribution, and overall cellular morphology may overlap significantly. Because of this similarity, distinguishing these two cell types correctly becomes difficult even for experienced pathologists. This may lead to false-positive diagnoses, unnecessary follow-up procedures, and increased patient anxiety.

Instead of working on a general five-class classification problem, this project mainly focuses on improving the distinction between Metaplastic and Koilocytotic cells, since they represent one of the most confusing cases in the SIPaKMeD dataset.

This study is designed as a supervised binary image classification problem using deep learning methods. The main goal is to automatically distinguish Metaplastic and Koilocytotic cervical cell images and improve the classification performance between these visually similar cell types. The model will learn texture and morphological differences directly from Pap smear images in order to make more accurate predictions.

In real clinical environments, pathologists need to examine a large number of microscopic cell images every day, which makes the screening process both difficult and time-consuming. Because of fatigue and subjective interpretation, diagnostic mistakes may sometimes occur, especially in visually similar cases. For this reason, the proposed system aims to support pathologists during cervical cancer screening and help reduce workload by providing more consistent classification results.

In addition, Grad-CAM visualizations can be used to better understand which regions of the cell images affect the model's decisions during classification. Deep learning models are often considered “black box” systems because it is usually difficult to understand how they make predictions. For this reason, Grad-CAM can help visualize the regions that the model focuses on while making a decision. This may help doctors and researchers better understand the behavior of the system and evaluate whether the model is paying attention to important areas such as the nucleus instead of unrelated background regions.

III. Dataset & Data Understanding

The SIPaKMeD dataset will be used for cervical cell classification. SIPaKMeD is a publicly available medical image dataset created for Pap smear image analysis and cervical cancer research. The dataset contains microscopic images of cervical cells taken from Pap smear tests for cervical cancer screening.

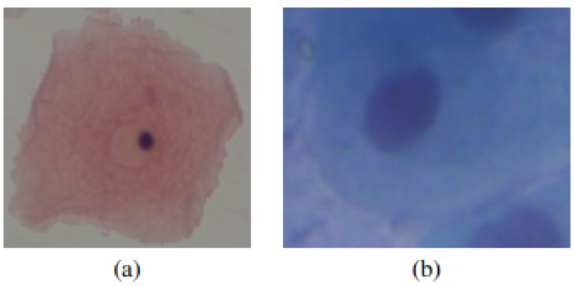

Fig. 1: Example image from the dataset

The original dataset includes approximately 4049 isolated cell images divided into five different classes:

- Parabasal (787 images)
- Superficial-Intermediate (831 images)
- Metaplastic (793 images)
- Koilocytotic (825 images)
- Dyskeratotic (813 images)

Although the dataset contains five classes, only Metaplastic and Koilocytotic classes will be used in this study.

Among these classes, Parabasal and Superficial-Intermediate cells are considered normal cell types. Koilocytotic and Dyskeratotic cells are considered abnormal because they are associated with precancerous changes. Metaplastic cells are benign, but they are visually very similar to abnormal Koilocytotic cells. This similarity makes the classification process more difficult and represents one of the main challenges of the project.

The images in the dataset are RGB microscopic images and contain different lighting conditions, staining differences, and cell appearances. Since the images come from real medical examinations, some image quality problems are naturally present. For example, some images contain small amounts of digital noise, slight blur, or illumination differences caused by the microscope and imaging process.

Another important issue is the dataset size. Although the dataset contains more than 4000 images, this is still relatively small for deep learning applications. Because of this, the model may memorize the training images instead of learning general features, which can lead to overfitting. To reduce this problem, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustment will be applied during training.

In addition, some classes contain more images than others, which may create a slight class imbalance problem. If this problem is not handled properly, the model may become biased toward classes with more samples. Data augmentation and cross-validation will therefore be used to improve generalization and create a more balanced training process.

Overall, the SIPaKMeD dataset provides a realistic and useful environment for developing and evaluating deep learning models for cervical cell classification.

IV. ML Pipeline Design

The pipeline for the morphological ambiguity in Pap smear slides will require mainly preprocessing to enhance features, data preparation and augmentation, feature extraction and model.

A. Pre-processing & Feature Enhancement

The images in the dataset contain impulse noise from the microscope image acquisition and have morphological ambiguity. To resolve this, we will use:

- **Median Filtering:** We will be denoising by using median filter while preserving cellular boundaries.
- **Contrast Limited Adaptive Histogram Equalization (CLAHE):** Enhancing local contrast and highlighting chromatin texture within the nucleus (found in cancerous cells) is important for distinguishing morphological differences.
- **Otsu-based Thresholding (Segmentation):** Segmenting the nucleus of the cell from the rest of the image is used to emphasize the nucleus region for this specific problem of distinguishing koilocytotic and metaplastic cells.

B. Data Preparation & Augmentation

All images need to be resized to a fixed resolution and normalized. Augmentation is required to handle the class imbalance such as rotating, scaling, and flipping.

C. Feature Extraction & Model

Feature extraction is performed automatically using a deep convolutional neural network (CNN) based on transfer learning. A pre-trained ResNet-50 model will be used as the backbone. Instead of relying on handcrafted features, the model learns hierarchical representations:

- Early layers extract low-level features such as edges and gradients
- Intermediate layers capture structural information such as cell boundaries and nucleus shape
- Deep layers learn high-level distinctions between Metaplastic and Koilocytotic cells

The final layer of the network will be modified for binary classification, using a single output neuron with sigmoid activation to predict the probability of a sample belonging to the Koilocytotic class. To adapt the model to domain-specific characteristics, selected deeper layers will be fine-tuned during training.

To improve model generalization and prevent overfitting:

- Dropout layers will be used in the classifier
- Early stopping will monitor validation loss
- The Adam optimizer will be employed with learning rate tuning

To ensure the model's decisions are clinically meaningful, Grad-CAM (Gradient-weighted Class Activation Mapping) will be used to visualize the regions influencing predictions.

This allows verification that the model focuses on the nucleus and chromatin structure, rather than background noise, improving trust in the system for medical personnel who may want to visually observe the model's decision-making.

V. Evaluation Strategy

In this project, different evaluation methods and performance metrics will be used to analyze the classification performance of the proposed model in a reliable way. Since the main objective of the study is to distinguish between Metaplastic and Koilocytotic cells, using only accuracy would not be sufficient for a complete evaluation. Therefore, several evaluation metrics will be considered together during the analysis process.

The dataset will first be divided into three main parts: training, validation, and test sets. Approximately 70% of the images will be used for training, 15% for validation, and the remaining 15% for testing. The training set will be used to train the model, the validation set will help monitor model performance during training, and the test set will be used for the final evaluation of the system.

In addition to the train-test split, 5-fold cross-validation will also be applied in order to obtain more reliable results. In this method, the dataset will be divided into five different parts, and the model will be trained and tested multiple times using different combinations of these parts. This approach helps reduce the dependency on a single data split and provides a better understanding of the model's generalization performance.

Several evaluation metrics will be used during testing. Accuracy will measure the overall classification performance of the model. In addition, precision, recall, and F1-score values will be used to evaluate the model more comprehensively. These metrics are especially important for analyzing the classification performance between Metaplastic and Koilocytotic cells, since these two classes are visually very similar.

A confusion matrix will also be generated to analyze classification errors in detail. By examining the confusion matrix, it will be possible to observe which cell types are more frequently misclassified by the model.

In addition, ROC curves and AUC values will be used to evaluate the model's ability to distinguish between the two cell classes at different threshold values. These metrics provide additional information about the discrimination capability of the proposed system.

Finally, the results obtained from different evaluation metrics will be analyzed together in order to provide a more comprehensive assessment of the proposed system. By combining accuracy, precision, recall, F1-score, confusion matrix analysis, and ROC-AUC evaluation, the overall classification capability and reliability of the model can be examined more effectively.

VI. Feasibility & Planning

This project is feasible because the SIPaKMeD dataset is public and already labeled. The dataset includes Pap smear cell images from five different classes. Because the images already have class labels, this project can be designed as a supervised binary image classification problem. Also, we do not need to collect new data or use special laboratory equipment. We will use Google Colab for coding and model training because it gives GPU support.

The project plan follows the official course schedule. In Week 10, the selected dataset was shared on the discussion forum. In Week 11, the project proposal report and proposal presentation will be prepared. In Week 12, the first progress report will be submitted. In this week, we plan to organize the dataset, check the class distribution, resize the images, and split the data into training, validation, and test sets. In Week 13, the second progress report will be submitted. In this week, we plan to apply data augmentation and train transfer learning models such as ResNet-50. In Week 14, the final report and final presentation will be prepared. The final results will be evaluated by using accuracy, precision, recall, F1-score, and confusion matrix.

The planned project schedule is shown in Table I.

TABLE I: Project Schedule

Week	**Planned Work**
Week 10	The SIPaKMeD dataset was selected and shared on the discussion forum.
Week 11	The project proposal report and proposal presentation will be prepared.
Week 12	The dataset will be organized. Images will be resized and split into training, validation, and test sets.
Week 13	Data augmentation will be applied. Transfer learning models will be trained and compared.
Week 14	The final model results, final report, and final presentation will be completed.

There are some possible risks in this project. The first risk is the small dataset size. A small dataset may cause overfitting. This means that the model may learn the training images very well but may not work well on new images. To reduce this risk, we will use data augmentation, dropout, early stopping, and transfer learning.

The second risk is class imbalance. One class may contain more samples than the other class. In this case, the model may learn the majority class better than the minority class. To reduce this problem, we will check the class distribution before training. If needed, we can use data augmentation or class weights.

The third risk is the visual similarity between some cell classes. For example, metaplastic and koilocytotic cells may look similar. This can make classification harder for the model. To understand this problem, we will use a confusion matrix. The confusion matrix will show whether Metaplastic and Koilocytotic cells are confused by the model.

Another risk is limited GPU time in Google Colab. Some deep learning models can take a long time to train. To reduce this risk, we will use lightweight transfer learning models such as ResNet-50. We will also use early stopping to avoid unnecessary training time.

Overall, this project is realistic and manageable within the course schedule. The dataset is available, the task is clear, and the planned method includes preprocessing, data augmentation, transfer learning, and evaluation metrics. These steps also satisfy the instructor's condition about using augmentation and transfer learning.

VII. Conclusion

This project proposes a deep learning-based system for cervical cell classification using the SIPaKMeD dataset. The main aim is to support Pap smear analysis and help classify cervical cell images more accurately. The project especially focuses on Metaplastic and Koilocytotic cells because these two cell types can look similar and may be difficult to distinguish.

The project will be designed as a supervised binary image classification task using only the Metaplastic and Koilocytotic classes from the original five-class SIPaKMeD dataset. Before training the model, the images will be resized, normalized, and divided into training, validation, and test sets. Since the dataset is not very large, data augmentation will be used to reduce overfitting and improve the model's performance on new images.

Transfer learning will be used in this project. A pre-trained model such as ResNet-50 will be used instead of training a model from the beginning. This method is useful because it can work better with a limited dataset. Dropout and early stopping will also be used to make the training process more stable.

The model will be evaluated using accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and AUC values. These metrics will help analyze the overall classification performance of the model and evaluate how successfully the system distinguishes between Metaplastic and Koilocytotic cells. The confusion matrix will be especially useful for checking the errors between Metaplastic and Koilocytotic cells.

Overall, this project is realistic and manageable within the course schedule. The dataset is available, the problem is clear, and the proposed method includes preprocessing, data augmentation, transfer learning, and proper evaluation metrics. Therefore, the project can be a useful support tool for cervical cell classification.

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